

《无穷级数》检测题参考解答

一、选择题 (15 分)

1. D 2. D 3. C 4. A 5. B

二、填空题 (15 分)

1. 8; 2. $\alpha > \frac{3}{2}$ 3. $1 < a \leq 3$ 4. $\begin{cases} 1, & x=0 \\ \frac{e^x-1}{x}, & x \neq 0 \end{cases}$ 5. $-\frac{1}{4}$

三、(8 分)

解 当 $p > 1$ 时, 取 $r > 0$, 且 $1 < r < p$, 则有 $\lim_{n \rightarrow \infty} \frac{\frac{\ln^q n}{n^p}}{\frac{1}{n^r}} = \lim_{n \rightarrow \infty} \frac{\ln^q n}{n^{p-r}} = 0$, 原级数收敛. (5分)

$\therefore$ 当 $p \leq 1$ 时, $\lim_{n \rightarrow \infty} \frac{\frac{\ln^q n}{n}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \ln^q n = +\infty$, $\therefore$ 原级数发散. (8分)

四、(8 分)

解 $\because \left| \frac{(-1)^n}{\sqrt{n+(-1)^n}} \right| = \frac{1}{\sqrt{n+(-1)^n}} \geq \frac{1}{\sqrt{n+1}}$, 而 $\sum_{n=2}^{\infty} \frac{1}{\sqrt{n+1}}$ 发散, $\therefore$ 原级数不绝对收敛, (3分)

$S_{2n} = \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{5}} - \frac{1}{\sqrt{4}} \right) + \cdots + \left(\frac{1}{\sqrt{2n+1}} - \frac{1}{\sqrt{2n}} \right)$, $\therefore S_{2n}$ 单调减少, (4分)

$S_{2n} > \left(\frac{1}{\sqrt{4}} - \frac{1}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{4}} \right) + \cdots + \left(\frac{1}{\sqrt{2n+2}} - \frac{1}{\sqrt{2n}} \right) = -\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2n+2}} > -\frac{1}{\sqrt{2}}$, (5分)

$\therefore \lim_{n \rightarrow \infty} S_{2n} = S$, 又 $\because \lim_{n \rightarrow \infty} S_{2n+1} = \lim_{n \rightarrow \infty} (S_{2n} + u_{2n+1}) = \lim_{n \rightarrow \infty} S_{2n} + \lim_{n \rightarrow \infty} \frac{1}{\sqrt{2n+2}} = S$, (7分)

$\therefore$ 原级数条件收敛. (8分)

五、(8 分)

解 $\because a_n > 0$, 且 $\{a_n\}$ 单调减少, $\therefore \lim_{n \rightarrow \infty} a_n = a \geq 0$, (2分) 若 $a = 0$, 则 $\sum_{n=1}^{\infty} (-1)^n a_n$ 收敛, 矛盾, (5分)

$\therefore a > 0$. $\because \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{1}{a_n+1} \right)^n} = \frac{1}{a+1} < 1$, (7分) $\therefore \sum_{n=1}^{\infty} \left(\frac{1}{a_n+1} \right)^n$ 收敛. (8分)

六、(8 分) 证 $\because f(-x) = f(x)$, $\therefore f'(0) = 0$, (1分)

由泰勒公式: $f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + o(x^2)$

令 $x = \frac{1}{n}$, $f\left(\frac{1}{n}\right) = 1 + \frac{1}{n^2} + o\left(\frac{1}{n^2}\right)$, $\Rightarrow f\left(\frac{1}{n}\right) - 1 = \frac{1}{n^2} + o\left(\frac{1}{n^2}\right) \sim \frac{1}{n^2}$, (4分)

$\Rightarrow \lim_{n \rightarrow \infty} \frac{\left| f\left(\frac{1}{n}\right) - 1 \right|}{\frac{1}{n^2}} = 1$, (7分) $\therefore \sum_{n=1}^{\infty} \left[f\left(\frac{1}{n}\right) - 1 \right]$ 绝对收敛. (8分)

七、(8分) 解 $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{2(n+1)-1}}{2(n+1)-1} \cdot \frac{2n-1}{x^{2n-1}} \right| = \lim_{n \rightarrow \infty} \frac{2n-1}{2n+1} x^2 = x^2,$

$\Rightarrow$ 当 $|x| < 1$ 时收敛, 当 $|x| > 1$ 时发散, $\therefore R = 1$, 当 $x = 1$ 时, $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1}$ 收敛,

当 $x = -1$ 时, $\sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1}$ 收敛, $\Rightarrow$ 收敛域: $[-1, 1]$, (2分)

$$S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots,$$

$$\text{则 } S'(x) = 1 - x^2 + x^4 - x^6 + \dots = \frac{1}{1+x^2}, \quad (4\text{分})$$

$$\int_0^x S'(x) dx = \int_0^x \frac{1}{1+x^2} dx = \arctan x,$$

$$\text{即 } S(x) - S(0) = \arctan x, \Rightarrow S(x) = \arctan x, \quad (7\text{分})$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} 3^{-\frac{2n-1}{2}} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} \left(\frac{1}{\sqrt{3}} \right)^{2n-1} = S\left(\frac{1}{\sqrt{3}} \right) = \arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}. \quad (8\text{分})$$

八、(10分) 解 $\because (\arctan x)' = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}, \quad x \in (-1, 1) \quad (2\text{分})$

$$\therefore \arctan x = \int_0^x (\arctan x)' dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1}, \quad x \in [-1, 1] \quad (4\text{分})$$

$$\begin{aligned} f(x) &= 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1} x^{2n} + \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+2} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n+1} x^{2n} + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n} \\ &= 1 + \sum_{n=1}^{\infty} \frac{(-1)^n 2}{1-4n^2} x^{2n}, \quad x \in [-1, 1] \quad (8\text{分}) \end{aligned}$$

$$\text{令 } x=1, \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{1-4n^2} = \frac{1}{2} [f(1) - 1] = \frac{\pi}{4} - \frac{1}{2}. \quad (10\text{分})$$

九、(10分) 解 将 $f(x)$ 作周期延拓, (1分)

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi-x}{2} \cos nx dx = 0 \quad (n=0, 1, 2, \dots) \quad b_n = \frac{1}{\pi} \int_0^{2\pi} \frac{\pi-x}{2} \sin nx dx = \frac{1}{n} \quad (n=1, 2, \dots) \quad (5\text{分})$$

$$\sum_{n=1}^{\infty} \frac{1}{n} \sin nx = \begin{cases} \frac{1}{2}(\pi-x) & 0 < x < 2\pi \\ 0 & x=0, 2\pi \end{cases} \quad (7\text{分})$$

$$\text{令 } x = \frac{\pi}{2}, \Rightarrow \frac{\pi}{4} = \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi}{2} = \sum_{n=1}^{\infty} \frac{\sin \left(\frac{2n-1}{2} \right) \pi}{2n-1} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{2n-1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}, \text{ 故 } \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \frac{\pi}{4}. \quad (10\text{分})$$

十、(10分) 解 (1) $\because \frac{1}{n}(a_n + a_{n+2}) = \frac{1}{n} \int_0^{\frac{\pi}{4}} \tan^n x (1 + \tan^2 x) dx = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}, \quad (3\text{分})$

$$(2) \because a_n = \int_0^{\frac{\pi}{4}} \tan^n x dx \geq 0, a_n + a_{n+2} = \frac{1}{n+1}, \therefore \frac{a_n}{n^\lambda} < \frac{a_n + a_{n+2}}{n^\lambda} = \frac{1}{n^\lambda(n+1)} < \frac{1}{n^{\lambda+1}}, \quad (7\text{分})$$

$\because \lambda > 0, \therefore \lambda + 1 > 1$, 故原级数收敛. (10分)